

9.2.32.

- a.** How many ways can the letters of the word ALGORITHM be arranged in a row?
- b.** How many ways can the letters of the word ALGORITHM be arranged in a row if A and L must remain together (in order) as a unit?
- c.** How many ways can the letters of the word ALGORITHM be arranged in a row if the letters GOR must remain together (in order) as a unit?
- a.** How many ways can the letters of the word ALGORITHM be arranged in a row?

A, L, G, O, R, I, T, H, M

The answer equals the number of 9-permutations of a set of nine elements, since there're nine letters of the word ALGORITHM, hence:

By definition of r-permutation and the given formula:

$$P(n, r) = \frac{n!}{(n-r)!}$$
$$P(9, 9) = \frac{9!}{(9-9)!} = \frac{9!}{0!} = \frac{9!}{1} = 9!$$

Solution: a. There're 9! ways to arrange ALGORITHM.

- b.** How many ways can the letters of the word ALGORITHM be arranged in a row if A and L must remain together (in order) as a unit?

AL, G, O, R, I, T, H, M

The answer equals the number of 8-permutations of a set of eight elements, since AL is treated as one object.

By definition of r-permutation and the given formula:

$$P(n, r) = \frac{n!}{(n-r)!}$$
$$P(8, 8) = \frac{8!}{(8-8)!} = \frac{8!}{0!} = \frac{8!}{1} = 8!$$

Solution: b. There're $8!$ ways to arrange ALGORITHM, where A and L must remain together (in order).

c. How many ways can the letters of the word ALGORITHM be arranged in a row if the letters GOR must remain together (in order) as a unit?

$A, L, \textcolor{teal}{GOR}, I, T, H, M$

The answer equals the number of 7-permutations of a set of seven elements, since GOR is treated as one object.

By definition of r-permutation and the given formula:

$$P(n, r) = \frac{n!}{(n-r)!}$$
$$P(7, 7) = \frac{7!}{(7-7)!} = \frac{7!}{0!} = \frac{7!}{1} = 7!$$

Solution: c. There're $7!$ ways to arrange ALGORITHM, where G, O and R must remain together (in order).